

Technical Appendix to

HIGHWAY TO SUCCESS: THE IMPACT OF THE GOLDEN QUADRILATERAL PROJECT FOR THE LOCATION AND PERFORMANCE OF INDIAN MANUFACTURING

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Appendix A. Empirical

This Appendix describes some key resources and data preparation steps regarding highway segments.

A.1. Data Source on GQ Details

National Highway Authority of India (NHAI), the GQ highway venture was a collection of 128 projects implemented all over the country. By March 2011, 120 of these projects had been completed, while eight were either pending completion or terminated. We compiled information on each of these 120 projects from the annual reports of NHAI from 1998–9 to 2010–11 as well as from the Ministry of Roads, Transport and Highways. These annual reports identified the project name for the highway stretch, the length of the highway stretch, the national highway number, the start date for the project and target and actual completion dates.

A.2. Mapping GQ Projects to Districts

In most cases, the name of the project indicated the start and end towns on a highway stretch. This information was used to identify districts lying within 10 kilometres of the highway stretch. The start and end points of the segment along the highway were located in the shape file and that segment was then selected to query all of the districts located within 10 kilometres of that segment in the GIS software. In some cases, the project name was not clear or the town name could not be located using the shape file, Google maps or open street maps. In such cases, we used information on the NHAI website for the highway project chainage and mapped the preceding or succeeding highway stretch. We then traced back kilometre by kilometre on the specified highway number to get the names of the towns that would lie close to the highway stretch.

A.3. New Construction versus Upgrades

We obtained information on whether each GQ project focused on new highway construction or the renovation and upgrade of an existing highway using a supplier database called Process Register. Process Register is a comprehensive online reference database of suppliers of products and services used in the process, energy and greater manufacturing industries. NHAI is a listed supplier on Process Register and most NHAI projects are listed in the database. Of the 70 districts lying near the GQ network, 37 districts experienced purely upgrade work. For 33 districts, some or all of the work was new construction.

A.4. *Implementation Date*

We grouped projects by whether they were completed by March 2003, March 2006 or later. We then matched districts to individual highway projects and their completion dates. Several districts touch two or more stretches of highway. For such districts, we allocated them to the earliest completion date as they had access to some connectivity before other districts did. However, we did not allocate a district into an earlier bin if the earlier GQ project was River Over Bridge (ROB), a bridge section, or a short bypass, as most of these constructions were small in terms of kilometres of length.

There were some highway projects, such as Vijayawada–Chilikaluripet, which came in several packages, and we could not distinguish among packages in terms of identifying districts because the start and end points were the same in all packages. Fortunately, in most cases, such projects were implemented in the same time period, so we could combine projects to get the set of districts lying within 10 kilometres of the project in question.

Of the 70 districts, 27 districts were completed prior to March 2003, 27 districts between March 2003 and March 2006 and 16 districts after March 2006. Two districts (Pali, Nadia) cannot be classified through the standard route and required research to ascertain. Pali was classified as being after March 2006 and new construction; Nadia was classified as being after March 2006 and upgrade. Additional details about these decisions are available from the authors.

A.5. *NS–EW Highway Phase I*

NS–EW highway projects were identified from the annual reports noted above. The reports do not explicitly identify projects on the NS–EW corridor as being Phase I or Phase II. Thus, a project was assigned to be part of Phase I if the target completion date was prior to December 2004, that is a year after the approval of the NHDP Phase II when a full-fledged NS–EW upgrade plan was approved by the Cabinet Committee on Economic Affairs. NS–EW corridor projects under NHDP Phase II did not start until January 2005. In total, 48 NS–EW projects were identified that aggregate approximately 981 kilometres of NS–EW planned to be completed in Phase I of NHDP. Of the 48 projects, 15 were to be implemented on the East–West corridor while the rest were scheduled for the North–South highway. Of the 76 districts lying within 0–10 kilometres of the NS–EW system, 40 districts were to be covered in the 48 NS–EW projects identified for Phase I.

Reference

- Sivadasan, J. (2009). 'Barriers to competition and productivity: evidence from India', *BE Journal of Economic Analysis & Policy*, vol. 9(1), Article 42.

Table A1
Descriptive Statistics for NS-EW Proximity

	Levels of total activity			Levels of young firm activity			Labour productivity (7)	Total factor productivity (8)
	Plants (1)	Employment (2)	Output (3)	Plants (4)	Employment (5)	Output (6)		
<i>(a) Average levels of activity in 1994 and 2000, combining districts within spatial range</i>								
Total	81,884	5,915,323	4.0E+11	12,035	556,463	4.5E+10	67,109	n.a.
Nodal district for NS-EW	8,303	516,706	3.6E+10	1,242	61,491	3.6E+09	68,827	0.028
District 0–10 km from NS-EW	21,070	1,299,014	8.0E+10	2,876	121,192	8.8E+09	61,304	–0.068
District 10–50 km from NS-EW	10,527	732,811	4.7E+10	1,690	78,331	5.5E+09	64,490	–0.114
District over 50 km from NS-EW	41,985	3,366,793	2.3E+11	6,227	295,448	2.7E+10	69,654	–0.091
<i>(b) Average levels of activity in 2005, 2007 and 2009 combining districts within spatial range</i>								
Total	95,678	7,621,581	8.1E+11	14,986	1,008,038	1.1E+11	106,385	n.a.
Nodal district for NS-EW	10,973	1,040,746	1.0E+11	1,833	151,191	1.3E+10	96,087	0.053
District 0–10 km from NS-EW	23,801	1,914,784	1.7E+11	3,198	212,260	2.0E+10	89,791	–0.143
District 10–50 km from NS-EW	11,178	781,925	7.8E+10	1,706	84,037	9.7E+09	99,891	–0.110
District over 50 km from NS-EW	49,726	3,884,127	4.6E+11	8,249	560,550	7.2E+10	118,631	–0.098
<i>(c) Ratio of activity in 2005/2007/2009 to 1994/2000 (Change for TFP)</i>								
Total	1.168	1.288	2.043	1.245	1.812	2.541	1.585	n.a.
Nodal district for NS-EW	1.322	2.014	2.812	1.476	2.459	3.476	1.396	0.025
District 0–10 km from NS-EW	1.130	1.474	2.159	1.112	1.751	2.229	1.465	–0.074
District 10–50 km from NS-EW	1.062	1.067	1.653	1.009	1.073	1.770	1.549	0.004
District over 50 km from NS-EW	1.184	1.154	1.965	1.325	1.897	2.674	1.703	–0.007
<i>(d) Change in share of activity between 2005/2007/2009 and 1994/2000</i>								
Nodal district for NS-EW	0.013	0.049	0.034	0.019	0.039	0.030	n.a.	n.a.
District 0–10 km from NS-EW	–0.009	0.032	0.011	–0.026	–0.007	–0.024		
District 10–50 km from NS-EW	–0.012	–0.021	–0.023	–0.027	–0.057	–0.037		
District over 50 km from NS-EW	0.007	–0.060	–0.022	0.033	0.025	0.031		

Note. See Table 1.

Table A2
Full Coefficient Set for Panel (b) of Table 2

DV: Change in manufacturing trait listed in column header	Log levels of total activity			Log levels of young firm activity			Total factor productivity	Log labour productivity	Log average wage	Log cost per employee
	Plants (1)	Employment (2)	Output (3)	Plants (4)	Employment (5)	Output (6)				
(0,1)Nodal district	0.541 (0.591)	0.468 (0.657)	0.493 (0.677)	0.831 (0.718)	0.964 (0.858)	0.927 (0.957)	1.367*** (0.280)	0.004 (0.151)	0.239*** (0.096)	0.249*** (0.100)
(0,1) District 0–10 km from GQ	0.312*** (0.124)	0.233* (0.129)	0.427*** (0.157)	0.616*** (0.174)	0.555*** (0.201)	0.680*** (0.286)	0.112 (0.215)	0.241*** (0.085)	0.169*** (0.060)	0.185*** (0.062)
(0,1) District 10–50 km from GQ	–0.117 (0.161)	–0.202 (0.196)	–0.024 (0.271)	–0.115 (0.207)	–0.025 (0.279)	–0.194 (0.416)	0.403 (0.288)	0.177 (0.127)	0.151* (0.087)	0.155* (0.090)
Log distance to national highway	–0.082 (0.051)	–0.049 (0.057)	–0.023 (0.070)	–0.041 (0.072)	–0.084 (0.088)	–0.007 (0.125)	–0.000 (0.087)	0.042 (0.038)	0.032 (0.025)	0.035 (0.026)
Log distance to state highway	0.017 (0.055)	0.041 (0.058)	–0.026 (0.066)	0.051 (0.074)	0.032 (0.091)	0.038 (0.125)	0.026 (0.096)	–0.049 (0.037)	–0.038 (0.024)	–0.036 (0.025)
Log distance to broad-gauge railroad	–0.005 (0.047)	0.018 (0.049)	0.111 (0.070)	0.000 (0.068)	0.018 (0.080)	0.063 (0.117)	0.156* (0.090)	0.087*** (0.044)	0.056*** (0.027)	0.054* (0.028)
Log total population	0.472*** (0.119)	0.424*** (0.124)	0.356*** (0.135)	0.531*** (0.175)	0.586*** (0.179)	0.837*** (0.233)	0.071 (0.146)	–0.038 (0.056)	–0.043 (0.042)	–0.051 (0.044)
Age profile/ demographic dividend	0.163* (0.093)	0.206* (0.108)	0.192 (0.129)	0.380*** (0.124)	0.596*** (0.161)	0.583*** (0.215)	–0.042 (0.167)	–0.042 (0.067)	–0.012 (0.046)	–0.016 (0.047)
Female–male sex ratio	–0.035 (0.076)	–0.062 (0.089)	–0.209* (0.108)	0.011 (0.106)	–0.159 (0.126)	–0.326* (0.172)	–0.103 (0.135)	–0.158*** (0.050)	–0.032 (0.036)	–0.021 (0.037)
Population share in urban areas	0.174** (0.084)	0.174* (0.093)	0.145 (0.110)	0.120 (0.130)	0.023 (0.164)	0.011 (0.205)	0.074 (0.129)	–0.017 (0.050)	0.084** (0.038)	0.098*** (0.039)
Population share in scheduled castes or tribes	0.113*** (0.049)	0.171*** (0.052)	0.180*** (0.077)	0.068 (0.069)	0.071 (0.086)	0.098 (0.132)	0.078 (0.115)	0.024 (0.061)	0.009 (0.033)	0.013 (0.034)
Literacy rate	–0.002 (0.079)	–0.057 (0.095)	0.081 (0.111)	–0.090 (0.110)	–0.138 (0.138)	0.045 (0.192)	0.445*** (0.160)	0.167** (0.078)	0.050 (0.050)	0.043 (0.052)
Infrastructure index	0.314*** (0.084)	0.390*** (0.093)	0.349*** (0.110)	0.142 (0.108)	0.056 (0.128)	0.161 (0.183)	0.090 (0.137)	0.007 (0.059)	0.008 (0.041)	0.016 (0.042)

Notes. See Table 2. Demographic variables are taken from the 2000 Population Census. Variables that are not expressed in logs or indicator variables are transformed to have unit standard deviation for interpretation.

Table A3
Further Robustness Checks on Panel (a) of Table 2

DV: Change in manufacturing trait listed in column header	Log levels of total activity			Log levels of young firm activity			Log labour productivity	Total factor productivity	Log average wage	Log cost per employee
	Plants (1)	Employment (2)	Output (3)	Plants (4)	Employment (5)	Output (6)				
<i>(a) Excluding sample weights and including outlier districts</i>										
(0,1) Nodal district	0.556 (0.593)	0.500 (0.661)	0.525 (0.679)	0.844 (0.718)	1.016 (0.861)	0.995 (0.960)	0.005 (0.154)	1.376*** (0.279)	0.239** (0.097)	0.250** (0.101)
(0,1) District 0–10 km from GQ	0.304** (0.124)	0.227* (0.130)	0.425*** (0.158)	0.609*** (0.175)	0.553*** (0.202)	0.679*** (0.289)	0.244*** (0.085)	0.108 (0.216)	0.167*** (0.061)	0.184*** (0.062)
(0,1) District 10–50 km from GQ	−0.121 (0.161)	−0.197 (0.197)	−0.019 (0.271)	−0.128 (0.205)	−0.027 (0.278)	−0.193 (0.417)	0.176 (0.126)	0.401 (0.289)	0.152* (0.086)	0.156* (0.090)
<i>(b) Using just the first distance band</i>										
(0,1) Nodal district	0.559 (0.592)	0.498 (0.659)	0.496 (0.675)	0.847 (0.717)	0.968 (0.856)	0.955 (0.957)	−0.023 (0.150)	1.306*** (0.275)	0.214** (0.095)	0.224** (0.099)
(0,1) District 0–10 km from GQ	0.334*** (0.119)	0.271** (0.127)	0.432*** (0.152)	0.638*** (0.167)	0.560*** (0.196)	0.717*** (0.277)	0.208*** (0.078)	0.037 (0.202)	0.139*** (0.057)	0.156*** (0.058)
<i>(c) Using finer distance bins within 10–50 km range</i>										
(0,1) Nodal district	0.552 (0.582)	0.503 (0.668)	0.506 (0.674)	0.825 (0.698)	0.960 (0.841)	0.936 (0.947)	−0.015 (0.151)	1.338*** (0.271)	0.225** (0.094)	0.235** (0.098)
(0,1) District 0–10 km from GQ	0.335*** (0.128)	0.256* (0.133)	0.450*** (0.159)	0.640*** (0.176)	0.582*** (0.204)	0.711** (0.288)	0.240*** (0.085)	0.113 (0.215)	0.168*** (0.060)	0.185*** (0.062)
(0,1) District 10–30 km from GQ	0.069 (0.291)	−0.148 (0.372)	0.008 (0.454)	0.226 (0.327)	0.215 (0.403)	0.138 (0.550)	0.125 (0.159)	0.170 (0.280)	0.089 (0.110)	0.091 (0.113)
(0,1) District 30–50 km from GQ	−0.056 (0.210)	−0.017 (0.192)	0.185 (0.316)	−0.198 (0.278)	0.026 (0.407)	−0.205 (0.615)	0.221 (0.172)	0.648 (0.435)	0.209* (0.120)	0.217* (0.125)
<i>(d) Using linear distance up to 50 km in non-nodal districts within 50 km of GQ network</i>										
Log distance from GQ	−0.134** (0.053)	−0.099* (0.059)	−0.081 (0.079)	−0.218*** (0.074)	−0.090 (0.103)	−0.116 (0.140)	0.006 (0.038)	0.097 (0.104)	0.025 (0.026)	0.023 (0.028)

Notes. See Table 2. Panel (d) uses information on 106 non-nodal districts within 50 kilometres of the GQ, with distances of less than one kilometre coded as being one kilometre from the GQ network.

Table A4
Alternative Definitions of District-level Labour and Total Factor Productivity

DV: Change in manufacturing trait listed in column header	Labour productivity		Sivadasan L-P approach		Residual approach	
	Weighted (1)	Unweighted (2)	Weighted (3)	Unweighted (4)	Weighted (5)	Unweighted (6)
(0,1) Nodal district	0.138 (0.111)	0.015 (0.010)	1.960*** (0.228)	1.971*** (0.195)	1.015*** (0.256)	1.228*** (0.280)
(0,1) District 0–10 km from GQ	0.199*** (0.074)	0.017*** (0.006)	0.220 (0.201)	0.163 (0.195)	0.334* (0.200)	0.355* (0.193)
(0,1) District 10–50 km from GQ	0.157 (0.126)	0.003 (0.011)	0.406 (0.302)	0.286 (0.280)	0.796** (0.363)	0.461 (0.308)

Notes. Robustness check on panel (a) of Table 2. TFP in columns (3)–(4) use Sivadasan (2009) approach to Levinsohn–Petrin estimations with repeated cross-section data. TFP in columns (5)–(6) use the average residual in each district from a weighted regression of log of value added on logs of employment and capital for each industry and year. Weighted estimations weight establishment-level TFPs by employment; unweighted estimations utilise simple averages.

Table A5
Negative Binomial Estimations on Entry Counts

DV: Levels of manufacturing trait listed in column header	Counts of total activity		Counts of young firm activity	
	Plants (1)	Employment (2)	Plants (3)	Employment (4)
(0,1) Post GQ upgrades ×	1.032***	1.019***	1.771***	2.716***
(0,1) District 0–10 km from GQ	(0.030)	(0.017)	(0.344)	(0.743)

Note. Robustness check on panel (a) of Table 6.

Table A6
Estimations of the Impact of GQ Improvements by Year, Districts within 50 kilometres of GQ Network

DV: Levels of manufacturing trait listed in column header	Log levels of total activity			Log levels of young firm activity			Log labour productivity	Total factor productivity	Log average wage	Log cost per employee
	Employment		Output	Employment		Output				
	Plants (1)	(2)	(3)	Plants (4)	(5)	(6)				
(0,1) Year 1999 × (0,1) District 0-10 km from GQ	0.035 (0.184)	0.188 (0.255)	0.359 (0.440)	-0.140 (0.265)	0.093 (0.376)	0.271 (0.625)	0.170 (0.223)	-0.148 (0.201)	0.128 (0.154)	0.166 (0.168)
(0,1) Year 2000 × (0,1) District 0-10 km from GQ	0.052 (0.187)	0.034 (0.168)	0.044 (0.225)	-0.048 (0.280)	0.216 (0.370)	0.359 (0.646)	0.016 (0.146)	-0.051 (0.157)	-0.075 (0.105)	-0.029 (0.113)
(0,1) Year 2001 × (0,1) District 0-10 km from GQ	-0.152 (0.210)	-0.147 (0.195)	-0.276 (0.249)	-0.108 (0.394)	0.086 (0.454)	0.172 (0.730)	-0.128 (0.176)	-0.168 (0.159)	-0.112 (0.105)	-0.072 (0.109)
(0,1) Year 2002 × (0,1) District 0-10 km from GQ	0.036 (0.202)	0.105 (0.201)	0.172 (0.268)	0.165 (0.334)	0.553 (0.357)	1.055* (0.539)	0.092 (0.138)	-0.062 (0.137)	0.037 (0.097)	0.088 (0.119)
(0,1) Year 2003 × (0,1) District 0-10 km from GQ	0.036 (0.211)	0.094 (0.240)	0.077 (0.297)	0.188 (0.331)	0.561 (0.384)	0.992* (0.534)	-0.019 (0.168)	0.128 (0.156)	-0.081 (0.122)	-0.037 (0.124)
(0,1) Year 2004 × (0,1) District 0-10 km from GQ	0.175 (0.139)	0.134 (0.157)	0.303 (0.276)	0.658** (0.306)	1.093*** (0.365)	1.785*** (0.499)	0.168 (0.190)	0.010 (0.156)	-0.020 (0.111)	0.034 (0.123)
(0,1) Year 2005 × (0,1) District 0-10 km from GQ	0.063 (0.142)	0.058 (0.147)	0.199 (0.198)	0.431 (0.306)	0.789* (0.417)	1.363** (0.543)	0.142 (0.117)	-0.016 (0.167)	-0.059 (0.089)	-0.002 (0.102)
(0,1) Year 2006 × (0,1) District 0-10 km from GQ	0.023 (0.150)	0.024 (0.129)	0.099 (0.173)	0.444 (0.297)	0.668* (0.383)	0.927* (0.479)	0.077 (0.117)	-0.037 (0.163)	-0.068 (0.083)	-0.065 (0.084)
(0,1) Year 2007 × (0,1) District 0-10 km from GQ	0.127 (0.142)	0.026 (0.136)	0.134 (0.172)	0.416 (0.283)	0.530 (0.427)	1.072* (0.547)	0.108 (0.125)	0.062 (0.130)	-0.030 (0.089)	0.027 (0.101)
(0,1) Year 2008 × (0,1) District 0-10 km from GQ	0.102 (0.156)	0.148 (0.158)	0.352* (0.194)	0.375 (0.282)	0.559 (0.346)	1.135** (0.475)	0.205 (0.134)	-0.044 (0.140)	-0.019 (0.097)	0.015 (0.108)
(0,1) Year 2009 × (0,1) District 0-10 km from GQ	0.338* (0.199)	0.399 (0.263)	0.553 (0.359)	0.905*** (0.290)	0.730** (0.352)	1.079** (0.480)	0.156 (0.169)	0.006 (0.157)	-0.095 (0.115)	-0.057 (0.125)

Notes: Dynamic estimation considers the location and productivity of organised-sector manufacturing activity in non-nodal Indian districts within 50 kilometres of the GQ network for 1994-2009 from the Annual Survey of Industries. The interaction terms quantify the differential effect for GQ upgrades for non-nodal districts within 10 kilometres of the GQ network compared to districts that are 10-50 kilometres from the GQ network by year, with 1994 as the reference year. The GQ upgrades commenced in 2001. Estimation reports standard errors clustered by district, include district and year fixed effects, have 1188 observations and weight observations by log total district population in 2001.

Table A7
Estimation of the Impact of GQ Improvements by Time Since Completion, Districts within 50 kilometres of GQ Network

DV: Levels of manufacturing trait listed in column header	Log levels of total activity			Log levels of young firm activity			Log labour productivity	Total factor productivity	Log average wage	Log cost per employee
	Plants (1)	Employment (2)	Output (3)	Plants (4)	Employment (5)	Output (6)				
Four years before upgrade in the district completes	0.083 (0.096)	-0.046 (0.130)	0.036 (0.186)	0.255 (0.191)	0.278 (0.219)	0.596** (0.292)	0.084 (0.098)	0.081 (0.115)	-0.080 (0.075)	-0.043 (0.083)
Three years before upgrade in the district completes	0.083 (0.085)	0.035 (0.111)	0.213 (0.132)	0.012 (0.171)	0.070 (0.208)	0.353 (0.329)	0.179** (0.077)	0.127 (0.101)	0.078 (0.052)	0.109* (0.057)
Two years before upgrade in the district completes	0.017 (0.097)	-0.017 (0.121)	0.118 (0.143)	0.152 (0.165)	0.197 (0.247)	0.568 (0.380)	0.135* (0.077)	-0.016 (0.123)	0.028 (0.058)	0.050 (0.063)
One year before upgrade in the district completes	0.174 (0.111)	0.088 (0.110)	0.278** (0.139)	0.344* (0.191)	0.239 (0.257)	0.648* (0.366)	0.192** (0.081)	0.056 (0.122)	0.032 (0.066)	0.054 (0.072)
Year upgrade completes in the district	0.179 (0.112)	0.069 (0.134)	0.225 (0.152)	0.352* (0.180)	0.218 (0.253)	0.434 (0.400)	0.157* (0.088)	0.120 (0.131)	0.001 (0.067)	0.034 (0.072)
One year after the upgrade in the district completes	0.125 (0.111)	-0.003 (0.122)	0.180 (0.154)	0.320 (0.194)	0.165 (0.257)	0.516 (0.396)	0.184** (0.091)	0.170 (0.114)	-0.039 (0.068)	-0.014 (0.074)
Two years after the upgrade in the district completes	0.127 (0.126)	0.003 (0.132)	0.133 (0.173)	0.546*** (0.204)	0.314 (0.289)	0.717* (0.416)	0.135 (0.098)	0.147 (0.118)	-0.047 (0.071)	-0.017 (0.074)
Three years after the upgrade in the district completes	0.268** (0.129)	0.151 (0.131)	0.263 (0.170)	0.597*** (0.205)	0.525* (0.292)	1.015** (0.447)	0.112 (0.103)	0.161 (0.118)	-0.064 (0.075)	-0.057 (0.078)
Four years after the upgrade in the district completes	0.256 (0.161)	0.128 (0.199)	0.285 (0.215)	0.644** (0.246)	0.591* (0.333)	1.065** (0.490)	0.167 (0.119)	0.124 (0.129)	-0.073 (0.081)	-0.065 (0.084)
Five years after the upgrade in the district completes	0.198 (0.161)	0.111 (0.157)	0.237 (0.195)	0.633** (0.266)	0.573 (0.371)	1.043* (0.531)	0.129 (0.117)	0.121 (0.141)	-0.060 (0.079)	-0.055 (0.082)
Six or more years after the upgrade in the district completes	0.299 (0.187)	0.279 (0.202)	0.422* (0.246)	1.007*** (0.266)	0.822*** (0.389)	1.212*** (0.554)	0.135 (0.126)	0.142 (0.146)	-0.059 (0.089)	-0.054 (0.091)

Notes. See Appendix Table 6. Effects are timed from the completion dates of each segment's upgrades.

Table A8
Industry-level Traits for India's Manufacturing Sector

	Total establishment counts, 2000	Land and building intensity, 2000
15 Food products and beverages	2,962,970	0.03
16 Tobacco products	2,062,543	0.03
17 Textiles	2,239,348	0.09
18 Wearing apparel; dressing and dyeing of fur	2,785,199	0.04
19 Leather tanning; luggage, handbags, footwear	171,759	0.05
20 Wood and wood products; straw and plaiting articles	2,720,752	0.04
21 Paper and paper products	90,214	0.07
22 Publishing, printing and media reproduction	144,293	0.04
23 Coke, refined petroleum and nuclear fuel	7,429	0.03
24 Chemicals and chemical products	216,410	0.06
25 Rubber and plastic products	95,352	0.06
26 Other non-metallic mineral products	784,551	0.09
27 Basic metals	43,127	0.05
28 Fabricated metal products, except machinery	640,256	0.04
29 Machinery and equipment, n.e.c.	171,138	0.05
30 Office, accounting and computing machinery	303	0.03
31 Electrical machinery and apparatus, n.e.c.	67,896	0.07
32 Radio, television and comm. equipment	7,589	0.05
33 Medical, precision and optical instruments, watches	9,190	0.07
34 Motor vehicles, trailers and semi-trailers	24,186	0.06
35 Other transport equipment	17,495	0.06
36 Furniture, manufacturing n.e.c.	1,255,784	0.04
Unweighted averages	750,808	0.05

Note. Descriptive statistics taken from Annual Survey of Industries.

Table A9
Table 10 with Alternative Industry Intensity Metrics

DV: Change in manufacturing trait listed in column header	Log new establishment counts by indicated industry intensity			Log new employment levels by indicated industry intensity			Log new output levels by indicated industry intensity		
	0-25th (1)	25th-50th (2)	>50th (3)	0-25th (4)	25th-50th (5)	>50th (6)	0-25th (7)	25th-50th (8)	>50th (9)
<i>(a) Using capital intensity to group industries</i>									
(0,1) Nodal district	1.945*** (0.338)	1.828*** (0.532)	0.876 (0.579)	2.903*** (0.457)	2.466*** (0.739)	0.994 (0.662)	3.258*** (0.517)	2.666*** (0.981)	1.803*** (0.849)
(0,1) District 0-10 km from GQ	0.496*** (0.161)	0.383*** (0.163)	0.852*** (0.183)	0.820*** (0.241)	0.496* (0.267)	0.939*** (0.224)	0.969*** (0.308)	0.576 (0.368)	1.585*** (0.315)
(0,1) District 10-50 km from GQ	-0.232 (0.207)	-0.010 (0.175)	-0.214 (0.218)	-0.124 (0.347)	0.013 (0.279)	-0.093 (0.298)	-0.193 (0.474)	0.039 (0.402)	-0.013 (0.448)
<i>(b) Using labor intensity to group industries</i>									
(0,1) Nodal district	1.185** (0.490)	2.186*** (0.510)	1.441*** (0.497)	1.615*** (0.435)	3.291*** (0.707)	1.749*** (0.590)	1.976*** (0.696)	3.505*** (0.862)	2.525*** (0.818)
(0,1) District 0-10 km from GQ	0.820*** (0.143)	0.491*** (0.148)	0.675*** (0.204)	1.025*** (0.205)	0.932*** (0.251)	0.746*** (0.257)	1.362*** (0.285)	1.063*** (0.330)	1.184*** (0.364)
(0,1) District 10-50 km from GQ	-0.269 (0.215)	-0.063 (0.154)	-0.263 (0.224)	-0.138 (0.319)	0.002 (0.278)	-0.148 (0.303)	-0.237 (0.456)	-0.024 (0.374)	-0.084 (0.464)
<i>(c) Using materials intensity to group industries</i>									
(0,1) Nodal district	1.740*** (0.464)	1.472*** (0.574)	1.441*** (0.522)	2.376*** (0.591)	2.081** (0.820)	1.916*** (0.494)	3.071*** (0.781)	2.467*** (1.002)	2.424*** (0.661)
(0,1) District 0-10 km from GQ	0.703*** (0.194)	0.623*** (0.173)	0.758*** (0.155)	0.845*** (0.244)	0.764*** (0.271)	0.992*** (0.217)	1.369*** (0.350)	1.056*** (0.377)	1.227*** (0.288)
(0,1) District 10-50 km from GQ	-0.202 (0.203)	-0.183 (0.209)	-0.213 (0.226)	-0.031 (0.278)	-0.154 (0.312)	-0.145 (0.352)	0.173 (0.423)	-0.200 (0.435)	-0.290 (0.480)

Note. See Table 10.

Table A10
Table 11 with Alternative Industry Intensity Metrics

DV: Change in manufacturing trait listed in column header	Log new establishment counts by indicated industry intensity			Log new employment levels by indicated industry intensity			Log new output levels by indicated industry intensity		
	0-25th (1)	25th-50th (2)	>50th (3)	0-25th (4)	25th-50th (5)	>50th (6)	0-25th (7)	25th-50th (8)	>50th (9)
<i>(a) Using capital intensity to group industries</i>									
(0,1) Nodal district	1.414** (0.603)	1.119 (0.706)	0.391 (0.879)	1.693* (0.875)	1.416 (1.034)	0.309 (1.105)	1.214 (1.001)	1.400 (1.397)	0.793 (1.373)
(0,1) District 0-10 km from GQ	0.159 (0.180)	0.272 (0.174)	0.494** (0.214)	0.361 (0.290)	0.318 (0.296)	0.352 (0.274)	0.336 (0.370)	0.318 (0.412)	1.013** (0.399)
(0,1) District 10-50 km from GQ	-0.285 (0.196)	0.057 (0.172)	-0.091 (0.202)	-0.255 (0.348)	0.100 (0.283)	-0.020 (0.284)	-0.496 (0.467)	0.079 (0.404)	0.195 (0.454)
<i>(b) Using labor intensity to group industries</i>									
(0,1) Nodal district	0.611 (0.715)	1.638** (0.702)	0.733 (0.763)	0.662 (0.819)	2.554** (1.054)	0.563 (1.042)	0.642 (1.153)	2.624** (1.196)	0.750 (1.382)
(0,1) District 0-10 km from GQ	0.482*** (0.180)	0.334** (0.157)	0.298 (0.227)	0.466* (0.265)	0.586** (0.282)	0.125 (0.295)	0.592 (0.374)	0.570 (0.381)	0.452 (0.428)
(0,1) District 10-50 km from GQ	-0.169 (0.197)	-0.028 (0.158)	-0.203 (0.212)	-0.131 (0.301)	0.076 (0.279)	-0.176 (0.294)	-0.301 (0.461)	0.045 (0.367)	-0.129 (0.445)
<i>(c) Using materials intensity to group industries</i>									
(0,1) Nodal district	1.298* (0.680)	0.683 (0.815)	0.890 (0.727)	1.542 (0.935)	1.103 (1.190)	0.597 (0.878)	1.709 (1.171)	1.231 (1.477)	0.626 (1.105)
(0,1) District 0-10 km from GQ	0.350* (0.207)	0.387** (0.184)	0.391** (0.198)	0.449 (0.294)	0.507* (0.295)	0.260 (0.276)	0.996** (0.422)	0.769* (0.415)	0.266 (0.367)
(0,1) District 10-50 km from GQ	-0.170 (0.188)	-0.038 (0.195)	-0.225 (0.202)	-0.009 (0.272)	0.004 (0.297)	-0.239 (0.329)	0.278 (0.400)	0.001 (0.420)	-0.505 (0.466)

Note. See Table 11.